

## 09. GENERALITIES CONTINUED

DURATION : 05:52

STUDY SHEET

### COURSE SUMMARY

This video covers the fundamental aspects of embryonic development, including the concept of the 'formative field' and the importance of energy without matter. Mechanisms such as mechanotransduction, polarity, and various environmental factors like heat and light are explored. You will learn how cell location influences their differentiation, as well as the differences between embryonic tissues, including ectodermal, endodermal, and mesodermal derivatives. The video concludes with an overview of the phases of embryogenesis, which are profoundly essential for any therapeutic practice.

## GENERALITIES ON EMBRYONIC DEVELOPMENT

The **potential for embryonic development** is comparable to a "**formative field**", an expression that describes a hypothetical field containing **energy without matter**. This concept is complex and includes elements such as nuclear impact, **tensegrity**, and the importance of the **matrix**.

We have examined mechanisms that go beyond a simple electromagnetic field. Factors such as **temperature**, **heat**, **mechanical stresses**, **hydrostatic pressures**, **electric fields**, and the **oxidation-reduction system** all play a crucial role in the system's **homeostasis**.

The presence of **elementary particles**, **electromagnetic energy**, the polarised human, the electromagnetic field, **Maxwell's law**, as well as the notion of **space** and **time** are also essential. **Light** is another important factor to consider.

### CELLULAR DIFFERENTIATION

How do a multitude of cells differentiate? We have observed that even minimal **changes** are linked to **mechanotransduction**.

A significant input of light modifies the parameters. When an electric field is present, there is always an electromagnetic field, which ensures the system's **cohesion** and provides a **reference**. Thus, a cell at a given location does not have the same spatial reference as another cell.

A cell's **genetic heritage** expresses itself differently at different times, even if all cells initially possess the same heritage. Cell layers will express themselves in different spaces and times, according to the **chronology** and **location** relative to the midline and the electric field.

The electromagnetic field provides the **information**. It is the position relative to this field that releases the information. Other parameters, such as **acidity**, are also involved, but **polarity** is the first major and very important component.

## ORIGIN AND LOCALISATION

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Where do **polarity**, **age**, and these forces come from? They existed long before us and will continue to exist after us.

The **energy of vibratory matter**, concentrated like light, induces matter, fluid, electricity, and **densification**, before dissipating. What makes one cell **hepatic** and another **cerebral** is its **localisation**. It is this localisation in space and time that determines differentiation.

Sometimes, cell clusters can be displaced and redifferentiate, but this depends on the timing of development.

## TISSUES AND EMBRYONIC DERIVATIVES

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You now have a good understanding of the difference between an **epithelial tissue** and a **connective tissue**. You can also distinguish between **ectodermal**, **endodermal**, and **mesodermal** tissue.

Let's take the **tongue**. It contains **muscle** (mesoderm) and is lined with epithelial tissue. It is a mixture of **endodermal** (epithelium) and **mesodermal** (muscle) tissues.

The **pancreas** is an endodermal derivative. All hormonal structures derive from epithelial tissue, also called **eminence tissue**. In other words, all **glands**, whether **exocrine** or **endocrine**, always originate from epithelial tissue, deriving from the ectoderm or endoderm, never from the mesoderm.

This is an important point to remember, now that we have clarified the origins. The **neurocranium**, **viscerocranium**, and **cranial base** represent a balance between the neurocranium and viscerocranium.

## PHASES OF EMBRYONIC DEVELOPMENT

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Tissue derivatives form from the first three weeks. By the beginning of the third week, the process is well underway. We can divide development into three main periods:

- The free period of **embryogenesis**.
- The free period of **organogenesis**.
- The free period of **periocephaly** (in two months).

Within two to three months, all **noble structures** are already formed. After that, it is mainly about **maturation** and **growth**.

It is essential to emphasise the notion of **growth** and the fact that **motility** is a movement always present in the body, defined by the developmental system. This movement becomes more complex in different situations.

Growth and the **autogenetic range** are fundamental processes.